

CASE 16

WISKOTT-ALDRICH SYNDROME

Role of the actin cytoskeleton in T-cell function.

Many functions of T cells require the directed reorganization of the cell's cytoskeleton, in particular the actin cytoskeleton. The eukaryotic cell cytoskeleton as a whole consists of actin filaments, microtubules, and intermediate filaments. It provides a framework for the internal structural organization of the cell and is also essential for cell movement, cell division, and many other cell functions. In T cells, as in other animal cells, the actin cytoskeleton is found mainly as a meshwork of actin filaments immediately underlying the plasma membrane (Fig. 16.1). The actin cytoskeleton is a dynamic structure and can undergo rapid reorganization by the depolymerization and repolymerization of actin filaments. As we shall see for Wiskott-Aldrich syndrome, an inability of T cells to reorganize their actin cytoskeleton when required has profound effects on their function and thus on immune function as a whole.

The functions of T cells in immune defense involve interactions with effector cells, such as B cells, dendritic cells, and infected target cells, that are initiated by direct cell-cell contact via cell-surface receptors. For example, helper T cells interact with B cells through cell-surface receptors to stimulate B-cell proliferation and the subsequent differentiation into antibody-producing plasma cells. T-cell-B-cell interactions are also involved in isotype switching and the generation of memory cells, whereas cytotoxic T-cell killing of virus-infected target cells also involves direct contact with the target cell. These interactions are accompanied by reorganization within the T-cell cortical actin cytoskeleton that, for example, focuses secreted T-cell products onto the target cell.

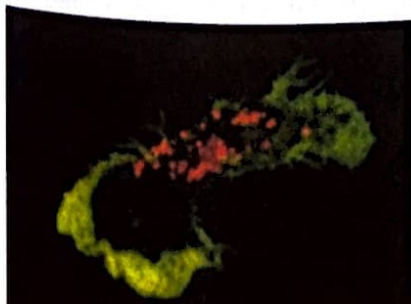


Fig. 16.1 An activated cytotoxic T cell moving over a surface. The actin cytoskeleton is stained green. The red staining indicates the lytic granules containing cytotoxic proteins. Courtesy of Gillian Griffiths.

TOPICS BEARING ON THIS CASE:

Isotype switching

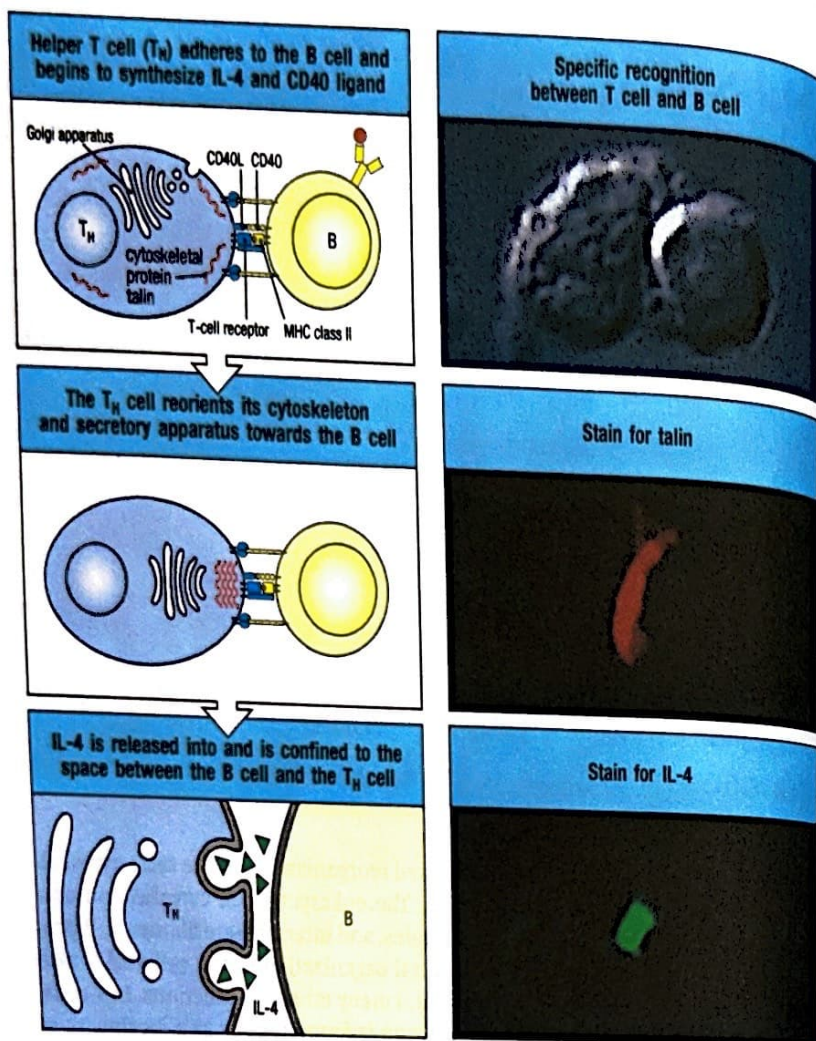
T-cell help in antibody responses

Interaction of cytotoxic T cells with their targets

Methods for measuring T-cell function

Gene knockouts

Fig. 16.2 Binding of a helper T cell to an antigen binding B cell causes a reorganization of the cytoskeleton in the T cell. Engagement of the T-cell receptor (TCR) by MHC class II / peptide complexes presented by B cells causes the T cell to adhere to the B cells via LFA-1 / ICAM-1 interactions and to express the CD40 ligand (CD40L), which binds to CD40 on the surface of the B cell (top panel). The cross-linking of the receptors at the point of contact leads to a reorganization of the cortical actin cytoskeleton, shown here by the redistribution of the protein talin (shown in red in top left and both middle panels), which is associated with the actin cytoskeleton. Subsequent reorientation of the secretory apparatus of the T cell leads to cytokine release at the point of contact with the antigen-binding B cell, as shown by staining for IL-4 (bottom right panel), which shows the IL-4 (green) confined to the space between the B cell and the T cell. Photographs courtesy of A. Kupfer.



The cytoskeleton is linked to cell-surface receptors in the plasma membrane so that events occurring at the membrane can drive cytoskeleton reorganization. For example, cross-linking of T-cell antigen receptors and co-receptors by antigen:MHC complexes leads to their aggregation at one pole of the T cell, with an accompanying concentration of the actin cytoskeleton at that point. Binding of a helper T cell to a B cell through the binding of its T-cell receptors to antigen:MHCII complexes on the B-cell surface also leads to a reorganization of the actin cytoskeleton locally in the zone of contact, which in turn causes a microtubule-dependent focusing of the secretory apparatus of the T cell on the point of contact with the B cell (Fig. 16.2); the release of cytokines from the T cell is thus directed to the contact point. Similar cytoskeletal reorganizations occur when a cytotoxic T cell recognizes antigen:MHC I complexes on its target cell.

Many other T-cell functions depend on the actin cytoskeleton. Like many animal cells, T cells move in a crawling 'ameboid' fashion. The movement of T cells as they emigrate from the thymus into the blood vessels and subsequently 'home' from the bloodstream into lymphoid tissue requires the active participation of the actin cytoskeleton. Cell division induced by the activation of T cells by antigen or by non-specific mitogens also involves the actin cytoskeleton, which, along with myosin, forms a contractile ring that divides the cell in two.

T cells from patients with Wiskott-Aldrich syndrome are deficient in all of these actin cytoskeleton-dependent functions and, in particular, seem unable to interact successfully with B cells and other target cells. In addition, because the Wiskott-Aldrich syndrome protein is broadly expressed within the hematopoietic system, other blood cell lineages are also affected.

The case of Jonathan Stilton: the consequences of a failure of T-cell-mediated immune responses.

Jonathan Stilton was first referred to the Children's Hospital at 20 months of age with a history of recurrent infections, eczema, asthma, and episodes of bloody diarrhea and autoimmune hemolytic anemia. He had been a full-term baby and seemed quite normal at birth. Routine immunizations of DPT (diphtheria, pertussis, and tetanus), conjugated *Haemophilus influenzae* type b (Hib) and pneumococcus vaccines, and inactivated polio vaccine had been given at 2, 4, and 6 months of age without any untoward consequences. At 6 months Jonathan's mother noticed eczema developing on his arms and legs; this was treated with 1% cortisone ointment. The eczema became infected with *Staphylococcus aureus*, and petechiae (small skin hemorrhages) appeared in the eczematous areas as well as on unaffected areas of skin. By 9 months of age, Jonathan began to have frequent infections of the middle ear and upper respiratory tract. At 1 year old, he developed pneumonia, which was confirmed by a chest radiograph. Between respiratory infections he started to wheeze and was found to have asthma.

At 16 months of age, Jonathan had an episode of bloody diarrhea. A blood analysis showed low levels of hemoglobin of 9.5 g dl^{-1} and a normal white blood cell count of $6750 \text{ } \mu\text{l}^{-1}$. The proportions of granulocytes and lymphocytes were also normal. The platelet count was, however, very low (thrombocytopenia), at $10,000 \text{ } \mu\text{l}^{-1}$ (normal $150,000\text{--}350,000 \text{ } \mu\text{l}^{-1}$), and the platelets were abnormally small (Fig. 16.3). He received two transfusions of platelets, after which the bloody diarrhea stopped, although the platelet count remained low at $40,000 \text{ } \mu\text{l}^{-1}$.

Two months later, Jonathan's mother noticed that he had become very pale, and his urine was darker than usual. He was brought in to the Emergency Room. At that time, his hemoglobin was as low as 5 g dl^{-1} and his platelet count remained low at $15,000 \text{ } \mu\text{l}^{-1}$. A positive Coombs direct antibody test was demonstrated, indicating that his red blood cells were coated with IgG antibody, and prompting the diagnosis of autoimmune hemolytic anemia. He was treated with corticosteroids (methylprednisone, 2 mg kg^{-1} body weight intravenously for 5 days). This resulted in an increase in hemoglobin levels to 8.5 g dl^{-1} . Jonathan was referred to the Children's Hospital for additional investigations.

Further immunological investigations at the time revealed IgG levels of 750 mg dl^{-1} (normal), IgM 25 mg dl^{-1} (decreased), IgA 475 mg dl^{-1} (increased), and IgE $1,750 \text{ ng l}^{-1}$ (increased). Jonathan's red blood cells were type O, but no anti-A or anti-B isohemagglutinins were present in his serum. His titer of antibodies against tetanus toxoid (a protein) was $0.15 \text{ unit ml}^{-1}$ (a borderline normal result). In addition, his antibody response to pneumococcus was protective to only two of the seven strains contained in the conjugated vaccine he had received. He was given a boosting immunization with polysaccharide vaccine to 23 pneumococcal strains, but 4 weeks later antibody titers remained unchanged.

Jonathan had normal numbers of circulating B cells (11% of peripheral blood lymphocytes) and T cells (85%) with a normal distribution of CD4 T cells and CD8 T cells. However, the *in vitro* proliferative response of his T cells to the mitogen phytohemagglutinin was slightly diminished, and response to the mitogenic effect of anti-CD3 antibodies was severely decreased.

Because of the combination of recurrent infections, eczema, thrombocytopenia with small platelets, bloody diarrhea, autoimmune hemolytic anemia, and antibody deficiency, Wiskott-Aldrich syndrome (WAS) was suspected. Flow cytometry and

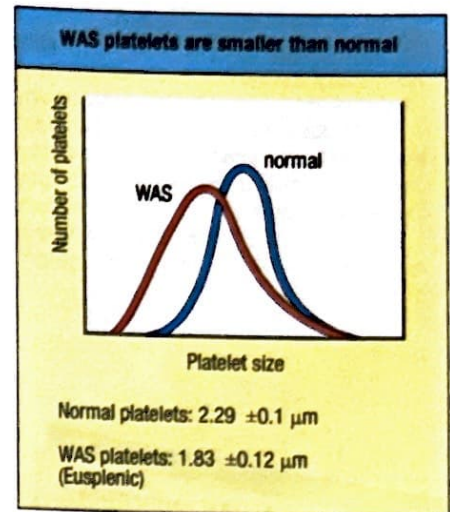


Fig. 16.3 Platelet sizing. Blood from a normal subject and from a patient with WAS (with a normal spleen) was drawn into sodium citrate, an anticoagulant. The blood was then centrifuged and allowed to sediment at $1g$; the platelet-rich plasma was removed. This was diluted in sterile buffer and scanned in a particle sizer. The mean diameter of normal platelets is $2.29 \pm 0.1 \text{ } \mu\text{m}$. The mean diameter of platelets from patients with Wiskott-Aldrich syndrome is $1.83 \pm 0.12 \text{ } \mu\text{m}$. Data kindly supplied by Dianne Kenney.

Normally developed male infant; recurrent infections, eczema, asthma, bloody diarrhea, autoimmune hemolytic anemia.

Thrombocytopenia with small platelets: Wiskott-Aldrich syndrome?

Poor antibody responses; reduced T-cell function.

Diagnosis of WAS confirmed;
severe phenotype. Consider
bone marrow transplantation.

Western blot analysis showed lack of expression of the Wiskott-Aldrich syndrome protein (WASP) in the cytoplasm of Jonathan's lymphocytes. A C→T change at position 71 in the WAS gene, resulting in a premature stop at codon 13, was demonstrated, confirming the diagnosis of WAS.

Treatment with intravenous immunoglobulins (400 mg kg⁻¹ every 3 weeks) and antibiotic prophylaxis with trimethoprim-sulfamethoxazole were started. Because of the severity of the disease, a bone marrow transplant was considered. Jonathan had two siblings, a 7-year-old brother and a 5-year-old sister. However, neither of them was HLA-identical. The search for a matched unrelated donor was initiated, and a suitable donor (matched at eight out of eight HLA loci tested) was identified 2 months later. At 26 months of age, Jonathan received a bone marrow transplant, following myeloablative chemotherapy. In addition, he received cyclosporin A for prophylaxis against graft-versus-host disease (GVHD). However, three weeks after transplant, he developed fever (at 39°C), skin rash, and slightly elevated liver enzymes (ALT 55 IU ml⁻¹; AST 47 IU ml⁻¹). Tests for pathogens were negative, and acute GVHD was diagnosed. Jonathan was treated with methylprednisone, resulting in rapid disappearance of the fever and rash, and normalization of liver enzymes. Now, several years after the bone marrow transplant, Jonathan lives at home. He has experienced no significant infections, his eczema has completely resolved and his platelet count has fully normalized at 220,000 ml⁻¹. He has received another round of immunization and has developed protective antibody titers to tetanus, to Hib, and to six out of the seven pneumococcal strains in conjugated anti-pneumococcal vaccine. His blood type has changed to B+ (the blood type of the bone marrow donor), and he has a normal titer of anti-A isohemagglutinin at 1:64.

Analysis of the family history revealed that Jonathan's mother had two older brothers. Of these, Mark was alive and well at 42 years of age, whereas Raymond had died at the age of 37 years. Raymond had developed eczema early in life that had persisted during the years. He had also suffered from recurrent bronchitis and had three episodes of pneumonia, which were treated with antibiotics. He also had petechiae and had been diagnosed with chronic idiopathic thrombocytopenic purpura because of his persistent thrombocytopenia. When he was 35 years old, Raymond developed a B-cell lymphoma positive for the Epstein-Barr virus. Although he was successfully treated with chemotherapy, Raymond died 2 years later of sepsis caused by a Gram-negative enterobacillus.

Wiskott-Aldrich syndrome.

Jonathan Stilton and his maternal uncle Raymond illustrate between them all the principal features of Wiskott-Aldrich syndrome (WAS). The syndrome was first described by Wiskott in 1937 in Munich when he observed three male infants with bloody diarrhea, eczema, and thrombocytopenia with small platelets. In 1947, Aldrich and his colleagues studied a large Dutch-American family in Minnesota and established the X-linked inheritance of the disease in five generations of affected males. The immunodeficiency that accompanies WAS was not appreciated until 1968, when it was described by Blaese and Waldmann and by Cooper and Good.

The WAS gene, mutations in which are responsible for WAS, maps to the short arm of the X chromosome. It encodes a protein named the Wiskott-Aldrich syndrome protein (WASP), which has homology with actin-binding cytoskeletal proteins involved in the reorganization of the actin cytoskeleton in white blood cells and platelets. WASP is expressed only in white blood cells and megakaryocytes (from which platelets are derived), which explains the restriction of its effects to immune system and blood clotting functions. In patients with WAS, T cells and

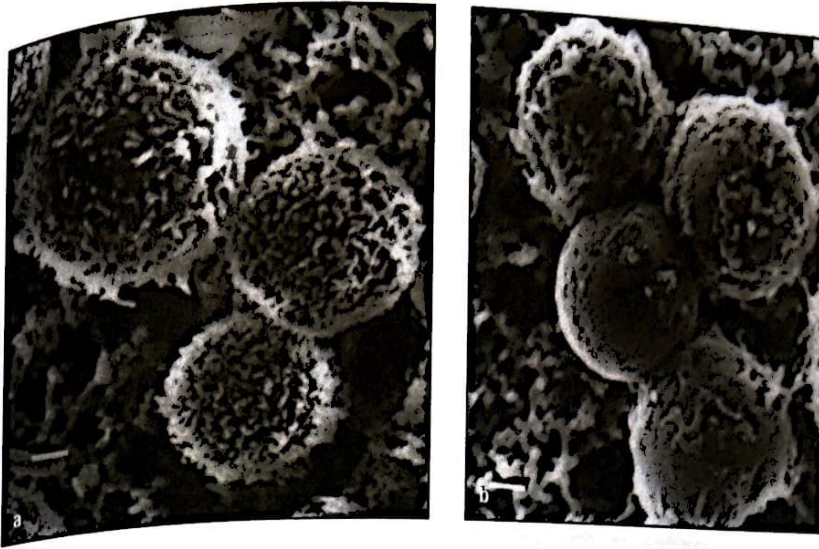


Fig. 16.4 Loss of microvilli on T cells in Wiskott-Aldrich syndrome. Scanning electron micrographs of normal lymphocytes (panel a) and lymphocytes from a patient with Wiskott-Aldrich syndrome (panel b). Note that the normal lymphocyte surface is covered with abundant microvilli, which are sparse or absent from the patient's lymphocytes. Photographs courtesy of Dianne Kenney.

platelets are defective in number and function. T cell movement, capacity for cell division, capping of antigen receptors, and reorientation of the cytoskeleton on engagement with other cells are all impaired. In addition, the function of monocytes, macrophages, and dendritic cells is also affected, with significant defects in directional motility and phagocytosis. The cytotoxic function of natural killer cells (NK cells) is impaired. Finally, distinct abnormalities of B lymphocytes have also been described.

Abnormally small platelets are a distinct and characteristic feature of WAS. The destruction of blood platelets by the spleen is greatly increased, but it is not known what the spleen recognizes as abnormal on WAS platelets. The platelets in the circulation become spontaneously activated and they extrude their granules. For this reason they appear small and their volume and diameter are smaller than normal (see Fig. 16.3). Splenectomy may increase the platelet number; however, it also increases the risk of sepsis and thus is not recommended. T cells are also morphologically abnormal in patients with WAS because the cells lose their surface microvilli and assume a characteristically bald appearance (Fig. 16.4).

Patients with WAS have increased susceptibility to both pyogenic bacterial infections and opportunistic infections. Among the latter, severe chickenpox (varicella), herpes simplex, and molluscum contagiosum (a viral infection of the skin, particularly eczematous skin) are frequently encountered. The increased susceptibility to such viral infections may be at least partly due to the impaired cytotoxic function of CD8⁺ T cells and NK cells in WAS; the impairment seems to be in their inability to attach to target cells. Antibody formation, particularly against carbohydrate antigens, is defective, as we saw for Jonathan, who could not respond to anti-pneumococcal vaccine. He also did not make antibodies against the blood group antigens, which are complex carbohydrates.

'Natural' antibodies of the IgM class include isohemagglutinins and antibodies directed against common blood-borne pathogens. The production of these natural antibodies as well as the production of antibodies to carbohydrate antigens in humans are dependent on the normal development of the marginal zone (MZ) in the spleen, which usually takes about 2 years to mature in normal individuals. Patients with WAS, however, have severe abnormalities of the splenic architecture, with virtual lack of the marginal zone and severely reduced numbers of IgM⁺CD27⁺ MZ B cells (Fig. 16.5).

Although Jonathan's T-cell number seemed normal when tested at 2 years old, patients with WAS typically show a progressive decline in T-cell count, often associated with poor control of viral infections, and especially of the Epstein-Barr virus (EBV), which can infect B lymphocytes. When this happens, the B cells undergo

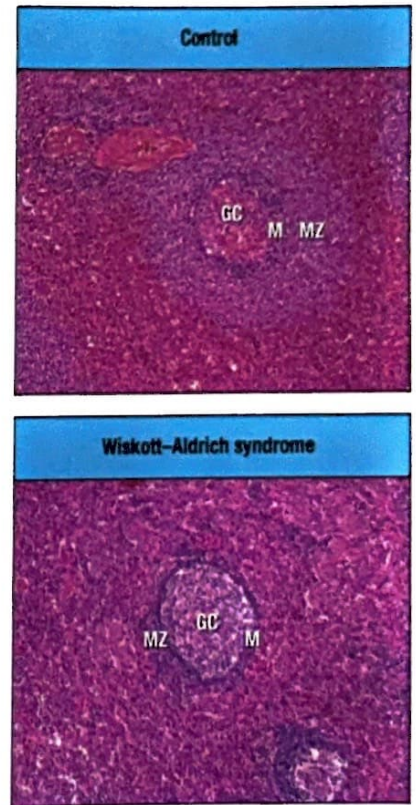
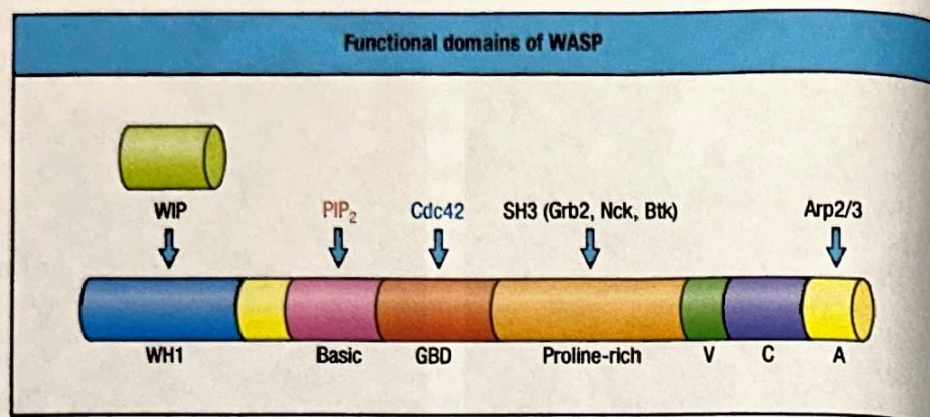


Fig. 16.5 Severe reduction of the marginal zone in the spleen from Wiskott-Aldrich syndrome patients. Hematoxylin and eosin staining of sections of spleen from a control subject (upper panel) and a patient with Wiskott-Aldrich syndrome (WAS) (lower panel). The germinal centers (GC) are surrounded by the mantle (M) and by the marginal zone (MZ), which contains B lymphocytes and specialized macrophages. In patients with WAS, the MZ is severely reduced in size. Magnification: $\times 40$. Photographs courtesy of Fabio Facchetti and William Vermi.

Fig. 16.6 A stick diagram of WASP (502 amino acids) shows the various domains of the molecule. The amino-terminal domain (WH1) binds WIP (the WASP interacting protein, which keeps WASP inactive until WIP is released). It is followed by a stretch of basic amino acids that bind the carboxy-terminal acid residues (A) of WASP, and this also keeps WASP in an inactive state. Phosphatidylinositol bisphosphate (PIP₂) also binds to this region and moves WASP from the cytoplasm to the immunologic synapse. Then comes a domain that binds the G protein Cdc42, followed by a proline-rich region, which is an SH3 domain that binds many adaptor proteins. Finally, the carboxy-terminal region of WASP is composed of verprolin (V)-like, cofilin (C)-like, and acidic (A) regions. These regions interact with the Arp2/3 complex and promote actin polymerization.



polyclonal expansion, which may eventually become oligoclonal or even monoclonal, leading to the development of B-cell lymphoma, as happened in one of Jonathan's uncles.

The structure and function of WASP have been studied intensively during the past few years (Fig. 16.6). The carboxy-terminal portion of the protein, the so-called VCA domain, binds to the actin-related protein complex (Arp2/3), and this binding initiates the nucleation of actin. Arp2/3 guides the branching of the actin filaments as they polymerize to make new filaments. WASP resides in the cytoplasm in an inactive form because the carboxy-terminal acidic domain (A) is bound to the GTPase-binding domain (GBD) so that inactive WASP appears as a hairpin. When the small G protein Cdc42 (in its GTP-bound form) binds to the GBD, WASP springs open into a linear activated form and the VCA domain can bind the Arp2/3 complex. Interestingly, missense mutations in the GBD domain of WASP result in a constitutively active form of the protein; this is associated with severe cellular abnormalities, especially in myeloid cells, that lead to severe congenital neutropenia and an increased risk of myelodysplasia, a pre-leukemic condition of the bone marrow. Finally, the amino-terminal WH1 domain of WASP binds to the WASP-interacting protein (WIP), and this binding stabilizes WASP. Amino acid substitutions in this domain result in decreased levels of WASP, and are typically responsible for a milder form of WAS, known as X-linked thrombocytopenia (XLT). Patients with XLT suffer from chronic thrombocytopenia, but they are less prone to severe infections, autoimmunity, or malignancies.

Mice with knockouts of the WAS gene show defective homing of lymphocytes and defective capping of the T-cell antigen receptor (Fig. 16.7), as well as defective responses of T cells to mitogens and anti-CD3 antibodies, defective formation of germinal centers, and severe abnormalities of marginal zone B cells.

The high frequency of immune-complex disease and allergy in WAS remains to be explained. It may relate to defective function of regulatory T cells. That may also account for the high frequency of autoimmune manifestations in patients with WAS, especially autoimmune cytopenias.

WAS is a severe disease. Although regular administration of immunoglobulins, antibiotic prophylaxis, and measures to avoid severe trauma-related bleeding are all important components of the treatment plan, severe manifestations and a failure to express WASP are an indication for hematopoietic cell transplantation. Excellent results have been achieved with transplantation from matched related or unrelated donors; however, prior myeloablative treatment is required to allow the robust engraftment of donor cells. Splenectomy is no longer considered a treatment of choice, because of the increased risk of sepsis. Initial trials of gene therapy to treat WAS are promising. Patients with XLT are typically treated with conservative management.

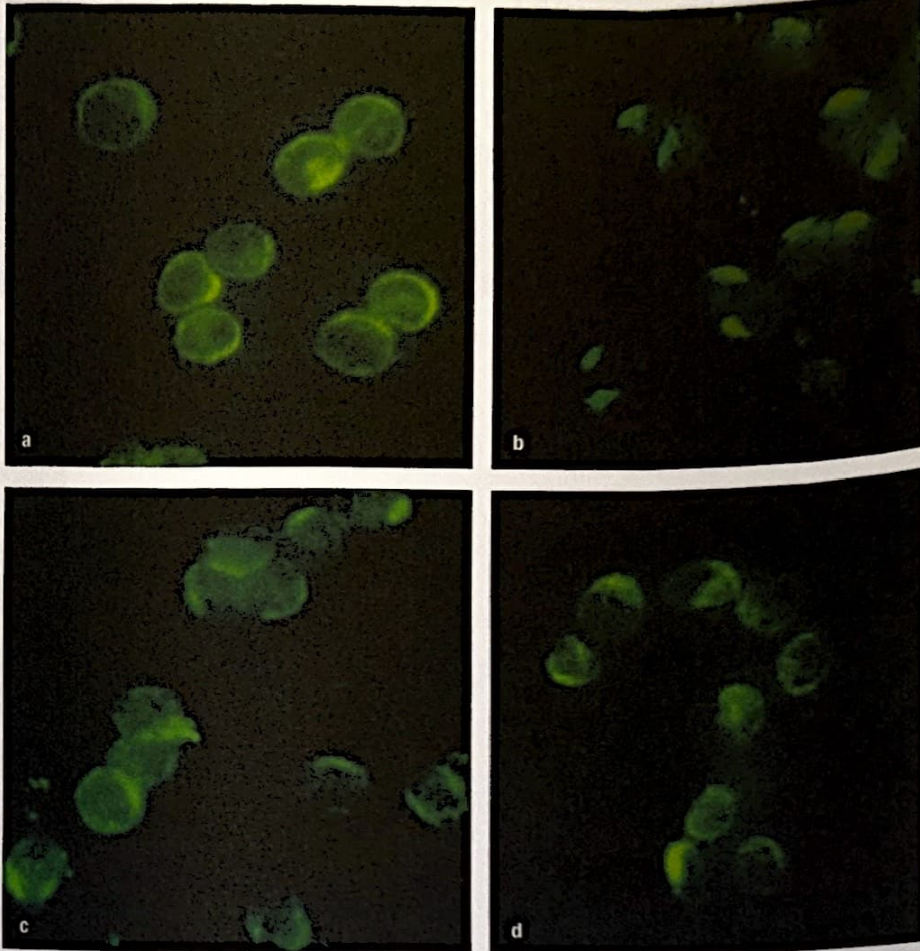


Fig. 16.7 Failure of T-cell capping in WAS knockout mice. T cells were obtained from $WAS^{-/-}$ mice in which the WAS gene encoding the protein WASP had been knocked out by homologous recombination. The T cells were treated with anti-CD3 ϵ antibody, which binds to the T-cell receptor:CD3 complex, and were then examined with a fluorescent antibody against CD3. Normal resting T cells (panel a) show capping of the T-cell receptor after anti-CD3 ϵ treatment (panel b). Panel c shows resting T cells from $WAS^{-/-}$ mice, and panel d shows the failure of cap formation after stimulation of the T-cell receptor. Photographs courtesy of Scott Snapper.

Questions.

- 1 When Mrs Stilton and other female heterozygous carriers of WAS are examined for randomness of X-chromosome inactivation, nonrandom inactivation of the WAS-bearing X chromosome is found in all the blood cell lineages—monocytes, eosinophils, basophils, neutrophils, B lymphocytes, and CD4 and CD8 T lymphocytes. When hematopoietic stem cells are isolated from these women, they also exhibit nonrandom X inactivation. How might this be explained?
- 2 Can you devise a strategy that might induce isotype switching in WAS B cells to overcome the lack of T-cell-B-cell collaboration?
- 3 Males with XLT are often initially given a diagnosis of chronic idiopathic thrombocytopenia purpura. Can you think of any laboratory test that would make you suspect XLT?